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Alignment Structure for a Fiber Optic Ferrule and Mechanical-Optical Interface

Abstract

A new fiber optic ferrule has an alignment structure on a surface through which light passes. The alignment structure is preferably in the shape of a dog bone, allowing the alignment structure to align the fiber optic ferrule in a receptacle while reducing the influence of temperature on the alignment.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application Ser. No. 62/656,894, filed on Apr. 12, 2018, and under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/046,956, filed on Oct. 12, 2020, and also to U.S. patent application Ser. No. 18/481,919, filed on Oct. 5, 2023, the contents of which are incorporated by reference.

FIELD OF THE INVENTION

[0002] Photonic-enabled transceiver devices typically are optically coupled to optical fibers by a fiber optic ferrule in which the optical fibers are secured. The fiber optic ferrule may have internal optical structures (e.g., lenses) that may be used to shape and/or direct the light beams between the transceiver and the optical fibers. The fiber optic ferrule may be coupled to a mechanical-optical interface (MOI) or other receptacle that may also have internal optical structures. In some instances, the other receptacle might be an adapter that is used to optically couple to fiber optic ferrules.

[0003] The transceiver devices and the fiber optic ferrules are typically aligned with one another to ensure proper transmission of the light between the components. The receptacle and the transceiver devices are usually aligned by active alignment. Once the receptacle and the transceiver device are aligned, then the fiber optic ferrule can be aligned with the receptacle and/or transceiver device. However, the devices are typically made from different materials, which have different coefficients of thermal expansion (CTE), and the different CTEs may affect the alignment of the various components. In general, a change in ambient temperature (ΔT) causes a change in a dimension (ΔD) of the material of the optical ferrule and/or lenses (expansion or contraction). These variables are related as: $\Delta D = D \times (\Delta T) \times \text{CTE}$, where D is a distance between the two CTE mismatched materials (e.g., lenses etched in Si and the alignment features made of ferrule plastic material). In the prior ferrules, this distance would correspond to the distance between guide pins used to align the fiber optic ferrules and the location of the light exiting/entering the fiber optic ferrule. In order to reduce the change in the dimensions (and thus the alignment), it would be better to have the alignment feature closer to the location of where the light exits/enters the fiber optic ferrule.

[0004] A fiber optic ferrule has been invented that reduces the distance between the alignment features and the exit/entry location of the light. The materials used to align the components have also been changed to materials with a similar CTE, reducing yet another variable in the equation.

SUMMARY OF THE INVENTION

[0005] The present invention is directed to a fiber optic ferrule that includes a main body having a plurality of optical fiber openings therein, each of the plurality of optical fiber openings configured to receive an optical fiber therein, a surface on the main body in optical alignment with the plurality of optical fiber openings, and an alignment structure extending from the surface outward away from the main body, the alignment structure having a first portion extending between two second portions along the surface, the second portions being on opposite sides of the first portion

[0006] In some embodiments, the fiber optic ferrule includes a light transmission window through which light passes, the light transmission window being disposed within the first portion of the alignment structure.

[0007] In some embodiments, the surface lies in a plane that is generally parallel to a longitudinal axis through the plurality of optical fiber openings.

[0008] In some embodiments, the surface lies in a plane that is generally perpendicular to a longitudinal axis through the plurality of optical fiber openings.

[0009] In another aspect, the invention is directed to a combination of a fiber optic ferrule and an optical component that includes a fiber optic ferrule that includes a main body having a plurality of optical fiber openings therein, each of the plurality of optical fiber openings configured to receive an optical fiber therein, a surface on the main body in optical alignment with the plurality of optical fiber openings, and a first alignment structure having a first portion extending between two second portions, the second portions being on opposite sides of the first portion, and the optical component has an engagement surface to engage a portion of the surface of the main body and a second alignment structure having a center portion extending between two outside portions, the outside portions being on opposite sides of the center portion, wherein one of the first and second alignment structures extend outward and the other of the first and second alignment structures extend inwardly.

[0010] Additional features and advantages of the invention will be set forth in the detailed description which follows, and in part will be readily apparent to those skilled in the art from that description or recognized by practicing the invention as described herein, including the detailed description which follows, the claims, as well as the appended drawings.

[0011] It is to be understood that both the foregoing general description and the following detailed description of the present embodiments of the invention, and are intended to provide an overview or framework for understanding the nature and character of the invention as it is claimed. The accompanying drawings are included to provide a further understanding of the invention, and are incorporated into and constitute a part of this specification. The drawings illustrate various embodiments of the invention, and together with the description serve to explain the principles and operations of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a perspective view of one embodiment of a fiber optic ferrule from the bottom according to the present invention;

[0013] FIG. 2 is a perspective view from the top of the fiber optic ferrule in FIG. 1;

[0014] FIG. 2A is side elevational view of the fiber optic ferrule in FIG. 2;

[0015] FIG. 3 is a perspective view of the fiber optic ferrule in FIG. 1 from the bottom and in alignment with a receptacle;

[0016] FIG. 4 is a perspective view of the fiber optic ferrule in FIG. 1 from the bottom and in alignment with a receptacle having lenses therein;

[0017] FIG. 5 is a perspective view of the receptacle in FIG. 4;

[0018] FIG. 6 is a plane view of a portion of the fiber optic ferrule and the receptacle in FIG. 3 showing the fit;

[0019] FIG. 7 is a perspective view of a second embodiment of a fiber optic ferrule from the front according to the present invention;

[0020] FIG. 8 is a side elevational view of the fiber optic ferrule in FIG. 7;

[0021] FIG. 9 is a front elevational view of the fiber optic ferrule in FIG. 7;

[0022] FIG. 10 is a perspective view of one embodiment of a receptacle to be used with the fiber optic ferrule in FIG. 9;

[0023] FIG. 11 is a perspective view of the fiber optic ferrule in FIG. 7, the receptacle in FIG. 10, a mechanical-optical interface, and circuit board;

[0024] FIG. 12 is a perspective view of two fiber optic ferrules of FIG. 7 and an adapter to align the two fiber optic ferrules; and

[0025] FIG. 13 is a perspective view of another embodiment of a fiber optic ferrule according to the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0026] Reference will now be made in detail to the present preferred embodiment(s) of the invention, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

[0027] One embodiment of a fiber optic ferrule **100** according to the present invention is illustrated in FIGS. 1-4. The fiber optic ferrule **100** has a main body **102** that extends between a front end **104** and a rear end **106**. A plurality of optical fibers **108** can be secured within the fiber optic ferrule **100**. The optical fibers **108** are inserted into the fiber optic ferrule **100** through a rear opening **110** through the rear end **106**. Each of the plurality of optical fibers **108** is inserted into one of a plurality of optical fiber openings **112** in a center portion of the main body **102**. The fiber optic ferrule **100** is a total internal reflection fiber optic ferrule meaning that the light to and from the optical fibers **108** encounters lenses **114** towards the front end **104** and passes through ferrule **100** through a bottom surface **116**. It will be understood by one of ordinary skill in the art in view of this patent application that instead of a single row of optical fibers **108**, two or more rows of optical fibers, or other arrangement of optical fibers **108** (instead of a row), may be used without departing from the scope of the present patent application.

[0028] With regard to the embodiment in FIGS. 1-4, Applicant notes that the term “front” or “forward” means that direction where the optical fibers are terminated, while the term “rear” or “rearward” is used to mean the direction from which the optical fibers enter into the fiber-optic ferrule or fiber optic connector. Thus, “front” is that part of the fiber optic ferrule **100** on the left side of FIG. 1 (see arrow) and “forward” is out and to the left. “Rear” is that part of the fiber optic ferrule that is on the right side of the page and “rearward” is toward the right.

[0029] The fiber optic ferrule **100** has two openings in a top surface **118** for an adhesive/epoxy to hold the plurality of optical fibers **108** within the fiber optic ferrule **100**. The first opening **120** is between the plurality of optical fiber openings **112** and the rear end **106**. The second opening **122** is between the plurality of optical fiber openings **112** and the front end **104**. The second opening **122** is usually in communication with the ends of each of the plurality of optical fibers **108**, but the plurality of optical fibers **108** may end before the second opening or even extend beyond the second opening **122**. See, e.g., U.S. Pat. Nos. 8,985,865; 9,618,771; and 9,983,365, owned by the same applicant as the present invention. However, the ferrule **100** need not have either or both of the openings **120,122** and may simply have only the opening at the rear end **106**.

[0030] There may also another opening or cavity in the main body **102** of the fiber optic ferrule **100**—lens opening **124**. This lens opening **124** has the plurality of lenses **114**, see FIG. 4, that may be exposed to air or covered with an anti-reflection coating, causing them to be total internal reflective lenses. As such, the light passing through the fiber optic ferrule **100** is turned 90 degrees within the fiber optic ferrule **100** by the plurality of lenses **114**. The plurality of lenses **114** may also condition the light in other ways, e.g., collimating the light, expanding or reducing the size of the light beam, etc.

[0031] On the bottom surface **116** is an alignment structure **130**. It is noted that the bottom surface **116** is also the surface through which the light passes to and from the plurality of optical fibers **108**. The alignment structure **130** extends from the bottom surface **116** and away from the main body **102**. In the middle **132** of the alignment structure is a light transmission window **134** that is in optical alignment with the plurality of lenses **114** and the plurality of optical fibers **108**. The light could from a photonic integrated circuit (PIC), a VCSEL, or from any another light source. Alternatively, light output from or input into the optical fibers may pass through lenses in another aperture in the MOI onto a photodetector or PIC of the transceiver below the lens array of the MOI. See FIGS. 4 and 5 and the related discussion below. It will be appreciated by one of ordinary skill in the art reading this patent application that while the term “transmission” is being used for

transmission window **134**, the discussion is equally applicable to reception of optical beams passing therethrough.

[0032] The alignment structure **130** is configured similar to the shape of a dog bone, the reasons for which will be discussed below. However, it should be noted that other configurations are possible. See, e.g., FIG. **13**. The alignment structure **130** has a first portion **136** extending between two second portions **138,140** along the bottom surface **116**. As illustrated, the first portion **136** is generally rectangularly shaped, but could have other configurations as long as the functions of the alignment structure **130** as noted below are satisfied. For example, the first portion **136** could have convex or concave sides rather than the straight ones. The two second portions **138,140** are illustrated as circular portions that connect with the first portion **136** at the ends thereof. The first portion **136** is preferably as long as the transmission window **134** to allow the light to pass therethrough, but it could be longer or shorter. The transmission window **134** can be longer than the first portion **136** so that the transmission window **134** extends into the second portions **138,140**. See, e.g., FIG. **3**. If the transmission window **134** is shorter than the first portion **136**, then the second portions **138,140** are closer to edges of the fiber optic ferrule **100**. In a variation of the alignment structure **130**, there may be more than one of each of the second portions **138,140**, without departing from the scope of this patent application. For example, on each side of the first portion **136**, there may be two circular portions that connect with the first portion **136** at the ends thereof.

[0033] The second portions **138,140** are generally circular as illustrated in the figures. The second portions **138,140** have a diameter that is about 0.766 mm, and the first portion **136** has a width of about 0.516 mm. Thus, the diameter of the second portions **138,140** is about 50% larger than the width of the first portion **136**. It is preferred that the diameter of the second portions **138,140** is at least 30% larger than the width of the first portion **136**. It is generally more preferable that the diameter of the second portions **138,140** is at least 50% larger than the width of the first portion **136**. It should also be noted that the second portions **138,140** may also have other configurations as long as the functions of the alignment structure **130** as noted below are satisfied. For example, the second portions **138,140** could be more oval, tear-drop shaped, or two rounded portions separated by a straight portion and still fall within the scope of the present invention. The first portion **136** may also have other features including, for example, a reduced width at the junction with the second portions **138,140** so the second portions **138,140** have more of a curved outer edge.

[0034] The bottom surface **116** lies in a first plane A, which is generally parallel to a plane B (and a longitudinal axis) in which the plurality of optical fibers **108** (and the plurality of optical fiber openings **112**) lie. However, it is possible that the planes A and B are not parallel to one another as illustrated in U.S. Pat. No. 8,985,865. The alignment structure **130** has a front face **142** that lies in a plane C that is also preferably parallel to planes A and B. See FIG. **2A**.

[0035] The alignment structure **130** extends a distance L of about 0.3 mm from the bottom surface **116** and away from the main body **102**. However, the distance that could be greater or less, depending on the need of the user and the structure that the fiber optic ferrule engages. The alignment structure **130** is preferably molded with the fiber optic ferrule **100** at the same time and from the same material as the fiber optic ferrule **100**. As such, the fiber optic ferrule **100** and the alignment structure **130** are integrally formed. Alternatively, the alignment structure **130** could be attached to the fiber optic ferrule **100** sometime after the fiber optic ferrule is molded. It could be made from the same material and attached in a number of different ways. It could be attached with an adhesive, such as epoxy, or ultrasonically welded to the main body **102**. Alternatively, the fiber optic ferrule **100** could be molded as a body with a larger thickness and then the alignment structure **130** is formed by removing some of the fiber optic ferrule material.

[0036] A receptacle **150** for receiving the fiber optic ferrule **100** is illustrated in FIG. **3** and a second receptacle **150'** is illustrated in FIGS. **4** and **5**. The only difference between the receptacle **150** and the receptacle **150'** is that receptacle **150'** has a plurality of lenses **154'** in the opening **152'**

that receives the alignment structure **130**. So for receptacle **150'** with a plurality of lenses **154'**, there may be no need to have the plurality of lenses in the main body **102**. The receptacle **150** (and receptacle **150'**) is preferably aligned with the transceiver on the board, but those structures are omitted from the drawings to increase clarity.

[0037] The fiber optic ferrule **100** with the alignment structure **130** can be aligned with the opening **152**. As noted above, the first portion **136** and second portions **138,140** align the fiber optic ferrule **100** with the receptacle **150** (and receptacle **150'**). In particular, the opening **150** has a first portion **154** and second portions **156,158**. However, the opening **150** is sized slightly different from the sizing of the alignment structure **130** and the first portion **136** and second portions **138,140**. The first portion **154** and second portions **156,158** of opening **150** are slightly larger than the first portion **136** and second portions **138,140**, respectively, but there may also be an interference fit. The difference in the sizes of the second portions is less than the difference in the sizes of the first portions because the second portions are the features that align the fiber optic ferrule **100** to the receptacle **150** (and receptacle **150'**). Indeed, it is the curved second portions **156,158** and **138,140** that control the alignment of the fiber optic ferrule **100** to the receptacle **150**. Again, the exact shape of the opening **150** can be changed to match the configuration of the alignment structure **130** and any changes to that configuration noted above.

[0038] Turning now to the embodiment in FIGS. 7-12, there is a different embodiment of a fiber optic ferrule **200**. Applicant notes that the term “front” or “forward” means that direction where the fiber optic connector would meet with another fiber optic connector or device, while the term “rear” or “rearward” is used to mean the direction from which the optical fibers enter into the fiber-optic ferrule or fiber optic connector. Thus, “front” is that part of the fiber optic ferrule **200** on the left side of FIG. 7 where the alignment structure **230** is located and “forward” is out and to the left. “Rear” is that part of the fiber optic ferrule that is on the right side of the page and “rearward” is toward the right. In FIG. 8, front and forward is to the right, and rear, rearward, and back are on the left side of the fiber optic ferrule **200**.

[0039] The fiber optic ferrule **200** has a main body **202** that extends between a front end **204** and a rear end **206**. A plurality of optical fibers (not shown but the same as optical fibers **108**) can be secured within the fiber optic ferrule **200**. The optical fibers are inserted into the fiber optic ferrule **200** through a rear opening **210** through the rear end **206**. Each of the plurality of optical fibers is inserted into one of a plurality of optical fiber openings **212** in a center portion of the main body **202**. See FIG. 12.

[0040] The fiber optic ferrule **200** has a front surface **216** through which the light to and from the optical fibers passes. Fiber optic ferrule **200** may have an opening **220** in a top surface **218** for an adhesive/epoxy to hold the plurality of optical fibers within the fiber optic ferrule **200**. See FIG. 12. It is the front surface **216** and the alignment structure **230** that will be the focus of the remainder of the discussion as the remainder of the fiber optic ferrule is known.

[0041] The alignment structure **230** extends from the front surface **216** and away from the main body **202**. In the middle **232** of the alignment structure are a plurality of lenses **214** that are in optical alignment with the plurality of optical fiber openings **212** (and the plurality of optical fibers).

[0042] The alignment structure **230** is configured similar to the shape of a dog bone, for the same reasons as noted above. As with the prior embodiment, other configurations are possible.

[0043] The alignment structure **230** has a first portion **236** extending between two second portions **238,240** along the front surface **216**. As illustrated, the first portion **236** is generally rectangularly shaped, but could have other configurations as long as the functions of the alignment structure **230** as noted above are satisfied. For example, the first portion **236** could have convex or concave sides rather than the straight ones. The two second portions **238,240** are illustrated as circular portions that connect with the first portion **236** at the ends thereof. The first portion **236** is preferably as long as the length of the plurality of lenses **214** to allow the light to pass therethrough, but it could be

longer or shorter.

[0044] Alternatively there could be a transmission window in the alignment structure **230** at the front surface **216** and the plurality of lenses **214** could be recessed or internal in the main body **202**.

[0045] The second portions **238,240** are generally circular as illustrated in the figures. The second portions **238,240** also have a diameter that is about 0.766 mm, and the first portion **236** has a width of about 0.516 mm. It is preferred that the diameter of the second portions **238,240** is at least 30% larger than the width of the first portion **236**. It is generally more preferable that the diameter of the second portions **238,240** is at least 50% larger than the width of the first portion **236**. It should also be noted that the second portions **238,240** may also have other configurations as long as the functions of the alignment structure **230** as noted below are satisfied. For example, the second portions **238,240** could be more oval, tear-drop shaped, or two rounded portions separated by a straight portion and still fall within the scope of the present invention. The first portion **236** may also have other features including, for example, a reduced width at the junction with the second portions **238,240** so the second portions **238,240** have more of a curved outer edge.

[0046] The front surface **116** lies in a first plane D, which is generally perpendicular to a plane B (and a longitudinal axis) in which the plurality of optical fiber openings **212** lie. The alignment structure **230** has a front face **242** that lies in a plane E that is also preferably parallel to plane D and perpendicular to plane B. See FIG. **8**.

[0047] The alignment structure **230** extends a distance L of about 0.3 mm from the front surface **216** and away from the main body **202**. However, the distance that could be greater or less, depending on the need of the user and the structure that the fiber optic ferrule engages. The alignment structure **230** is preferably molded with the fiber optic ferrule **200** at the same time and from the same material as the fiber optic ferrule **200**. As such, the fiber optic ferrule **200** and the alignment structure **230** are integrally formed. Alternatively, the alignment structure **230** could be attached to the fiber optic ferrule **200** sometime after the fiber optic ferrule is molded. It could be made from the same material and attached in a number of different ways. It could be attached with an adhesive, such as epoxy, or ultrasonically welded to the main body **202**. Alternatively, the fiber optic ferrule **200** could be molded as a body with a larger length and then the alignment structure **230** is formed by removing some of the fiber optic ferrule material.

[0048] A receptacle **250** with an opening **252** for receiving the fiber optic ferrule **200** is illustrated in FIGS. **10** and **11**. The function and design of the receptacle **250** is the same as described above and will not be repeated here. The fiber optic ferrule **200** is illustrated as being inserted into the receptacle **250** in FIG. **11** that also has an MOI **260** and a board **262**.

[0049] FIG. **12** illustrates two fiber optic ferrules **200** that will mate in an adapter **300**. The adapter **300** has an opening **302** that is the same as openings **150** and **250** to receive the alignment structure **230** on the fiber optic ferrules **200**.

[0050] Another embodiment of a fiber optic ferrule **400** according to the present invention is illustrated in FIG. **13**. The difference between the fiber optic ferrule **400** and the other fiber optic connectors is the shape of the alignment structure **430** on the front face **416**. The positioning, forming, and depth of the alignment structure **430** is also the same as noted above. The alignment structure **430** has a first portion **436** extending across the front face **416** and between two second portions **438,440**. The second portions **438,440** are on opposite sides of the width of the alignment structure **430** rather than the length thereof. This position moves the two second portions **438,440** closer to the center of the fiber optic ferrule **400** and may further reduce the effect of temperature changes on the alignment of the fiber optic ferrule **400**. In a variation of the alignment structure **430**, there may be more than one of each of the second portions **438,440**, without departing from the scope of this patent application. For example, on each side of the first portion **436**, there may be two circular portions that connect with the first portion **436**. Furthermore, the embodiment in FIG. **13** may be combined with the prior embodiments, and dimensions of any portion of the alignment structures **130**, **230**, **430** may be varied in such a combination to achieve optimum results.

[0051] It will be apparent to those skilled in the art that various modifications and variations can be made to the present invention without departing from the spirit and scope of the invention. Thus it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. A fiber optic ferrule comprising: a main body having a plurality of optical fiber receiving members therein to receive a plurality of optical fibers and arranged to form at least one row of optical fibers, each of the plurality of optical fiber receiving members configured to respectively receive an optical fiber in the plurality of optical fibers therein, each optical fiber to transport a light beam thereby forming at least one row of light beams; a surface on the main body optically aligned with the plurality of optical fiber receiving members, ends of the plurality of optical fiber receiving members being separated from the surface; and an alignment structure molded integrally with the fiber optic ferrule and having a first portion and a second portion, wherein the first portion is located externally on one side of the main body and the second portion is located externally on an opposite side of the main body.
2. The fiber optic ferrule according to claim 1, further comprising a light transmission window, the light transmission window being disposed between the first portion and the second portion of the alignment structure.
3. The fiber optic ferrule according to claim 1, further comprising a plurality of lenses, each of the plurality of lenses being in optical alignment with one of the plurality of optical fiber receiving members.
4. The fiber optic ferrule according to claim 3, wherein each of the plurality of lenses is disposed between the first portion and the second portion of the alignment structure.
5. The fiber optic ferrule according to claim 1, wherein the surface lies in a plane that is generally parallel to a longitudinal axis through the plurality of optical fiber receiving members.
6. The fiber optic ferrule according to claim 1, wherein the surface lies in a plane that is generally perpendicular to a longitudinal axis through the plurality of optical fiber receiving members.
7. The fiber optic ferrule according to claim 1, wherein the alignment structure is integrally formed with the main body of the fiber optic ferrule by a removal of ferrule material from the fiber optic ferrule.
8. The fiber optic ferrule according to claim 1, wherein the alignment structure is matingly receivable in a matching structure to position the fiber optic ferrule for mating.
9. The fiber optic ferrule according to claim 1, wherein the at least one row of light beams undergo total internal reflection at the fiber optic ferrule and pass through the surface substantially perpendicular to the plurality of optical fiber receiving members and the at least one row of optical fibers.
10. The fiber optic ferrule according to claim 1, wherein the at least one row of light beams pass through the surface in alignment with the plurality of optical fiber receiving members and the at least one row of optical fibers.
11. The fiber optic ferrule according to claim 1, wherein the fiber optic ferrule is a single-piece ferrule.
12. The fiber optic ferrule according to claim 1, wherein each of the first portion and the second portion have a non-linear shape.
13. A fiber optic ferrule comprising: a main body having a plurality of optical fiber receiving members therein to receive a plurality of optical fibers arranged to form at least one row of optical fibers, each of the plurality of optical fiber receiving members configured to respectively receive an optical fiber in the plurality of optical fibers therein, each optical fiber configured to transport a light beam thereby forming at least one row of light beams; a surface on the main body through

which the at least one row of light beams travels outside of the plurality of optical fibers; a first portion for an alignment of the fiber optic ferrule located externally on one side of the main body; and a second portion for an alignment of the fiber optic ferrule located externally on an opposite side of the main body, wherein the first portion and the second portion each being molded integrally with the fiber optic ferrule.

14. The fiber optic ferrule according to claim 13, wherein the main body extends along a longitudinal axis and the first portion and the second portion also extend substantially parallel to the longitudinal axis.

15. A fiber optic ferrule comprising: a main body with a top surface and a bottom surface having a plurality of optical fiber receiving members therein to respectively receive a plurality of optical fibers arranged in at least one row of optical fibers, each optical fiber transporting a light beam thereby forming at least one row of light beams; a surface on the main body through which the at least one row of light beams travels outside of the plurality of optical fibers, ends of the plurality of optical fiber receiving members being separated from the surface; and an alignment structure molded integrally with the fiber optic ferrule and having a first portion and a second portion on opposite sides of the at least one row of optical fibers and on opposite sides of the surface through which the at least one row of light beams travels, wherein the first portion and the second portion each have a non-linear shape between the top surface and the bottom surface.

16. The fiber optic ferrule according to claim 15, wherein the first portion and the second portion have a curvature.

17. The fiber optic ferrule according to claim 15, wherein the first portion and the second portion are integrally formed by a removal of a ferrule material from the fiber optic ferrule, thereby extending inwardly towards the main body of the fiber optic ferrule.

18. The fiber optic ferrule according to claim 15, wherein the fiber optic ferrule is a lensed ferrule.

19. The fiber optic ferrule according to claim 15, further comprising a substantially rectangular portion formed between the first portion and the second portion on the surface through which the at least one row of light beams pass.

20. The fiber optic ferrule according to claim 19, wherein the substantially rectangular portion is also integrally molded with the fiber optic ferrule.
